

Introduction

Amazon review data offers a public record of user engagement with products over time. By treating reviews as a proxy for product interaction, I will analyze which product categories are strong entry points for users, which categories are associated with repeat engagement, and which early product paths lead users into broader downstream category exploration.

Objective

Using a subset of the [UCSD Amazon reviews dataset](#), I will build a temporal user-product interaction model to analyze which product categories are strongest at retaining user engagement and which entry categories are most associated with later expansion into other high-value categories. The goal is to frame the analysis as an exploration of the data to support product growth opportunities.

Business / Product Questions

This project is designed to answer business-facing questions such as:

- Which product categories appear to be the highest-value candidates for further strategic investment?
- Which product categories are the strongest user entry points?
- Which categories attract many first-time reviewers, but fail to retain them?
- Which first reviewed categories are most associated with later expansion into high-retention categories?

Operationalizing Retention and Expansion

Retention

A user will be considered retained in a product category if, within **90 days** of their first review in that category, they leave **2 or more reviews on at least 2 distinct days** for products in the same category.

Retention rate

$\Sigma (\text{users retained in a category}) / \Sigma (\text{all users who reviewed in category})$

High-retention category

A high-retention category is a product category whose retention rate falls in the top quartile of observed category retention rates, among categories with at least a minimum number of entering users.

Expansion pathway

An entry category will be considered a strong expansion pathway if users whose first reviewed category is A have an above-baseline probability of later reviewing a product in a different, high-retention category within **90 days**.

ExpansionDifference_{A → B} = (

A = entry category,

B = destination category, where $B \neq A$, and *B* is high retention,

$P(B \text{ within } 90d \mid \text{first category} = A) - P(B \text{ within } 90d \mid \text{first category} \neq A)$

)

Raw Data Constraints:

Time range: May 1996 - September 2023

Product categories: 33 total for 2023 dataset

Data Subset

Time range: January 2023 to June 2023

Product Categories:

Electronics, Video Games, Software, and Cell Phones and Accessories

2 jsonl files per category:

- Review data
- Item metadata

Data Structure

User-product interaction layer

- **Entities (nodes):** Users and products
- **Interaction (edges):** A review connecting a user to a product
 - **Attributes:** timestamp, rating, category, brand, and other available product metadata
- **Purpose:** Used to identify first product/category entry, repeat engagement, and downstream movement across categories

Product transition layer

- **Entities:** Products or categories
- **Links:** Products/categories connected when the same user reviews both over time
- **Purpose:** Used to identify common customer movement patterns, entry points, and downstream expansion paths across product categories

Dream Scope

If I had unlimited time and resources, I would build an interactive product analytics tool that allows a user to search products or categories, inspect retention and expansion metrics, and trace common customer paths across all product categories. The application would help product, growth, or category-strategy teams identify where to improve onboarding, discovery, and cross-category recommendations.

Realistic Scope

By the deadline, I plan to complete a minimum viable version using a subset of the Amazon reviews dataset. I will clean and structure the review data, identify entry categories, calculate retention rates, and measure simple cross-category expansion patterns. If time permits, I will also add product-to-product transition analysis as a stretch goal.

Target grade: A-